Sinian Zhang

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EDUCATION

University of Minnesota, Twin Cities *Minneapolis, US*
Ph.D. Student in Biostatistics Sept 2024 – Jun 2029 (Expected)

Dissertation Advisors: Sandra Safo and Jue Hou (Division of Biostatistics and Health Data Science)
Also Mentored by: Ju Sun (Computer Science & Engineering)

Renmin University of China *Beijing, China*
B.S. in Statistics Sept 2020 – Jun 2024

RESEARCH INTERESTS

Statistically principled methods for **transferable and trustworthy machine learning and deep learning**, with applications spanning biomedical, imaging, and engineering domains.

- **Transfer and causal learning** – building estimators that keep their statistical guarantees when the data distribution shifts or when evidence must be pooled across sites, spanning distributional transfer learning, federated causal estimation under high-dimensional confounding, and reinforcement learning with generalized linear structure.
- **Uncertainty and selective prediction** – deciding when a model should abstain rather than guess, by designing selective prediction and classification rules that make the risk–coverage trade-off explicit and by quantifying the uncertainty those decisions rest on.
- **Agentic AI for scientific data** – multi-agent LLM systems that automate end-to-end analysis under human review, applied together with the methods above to electronic health records, biomedical knowledge graphs, and medical imaging.

PUBLICATIONS

Equal contribution.

Peer-Reviewed Publications

1. Elzbieta Jodlowska-Siewert, Yunhui Chen, **Sinian Zhang**, Jia Li, Robert Dellavalle, Rui Zhang, Jue Hou. Antidiabetic Drug Associations With Heart Failure Outcomes: Real-World Evidence Study Using Electronic Health Records. *JMIR Diabetes* 11 (2026): e85083.
   Role: Performed the electronic health record data processing.
2. Kaicheng Zhang*, **Sinian Zhang***, Doudou Zhou, Yidong Zhou. Wasserstein Transfer Learning. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
   Role: Co-developed the transfer-learning method and theory; led the simulation and real-data studies.
3. Yongkang Xiao, **Sinian Zhang**, Yi Dai, Huixue Zhou, Jue Hou, Jie Ding, Rui Zhang. DrKGC: Dynamic Subgraph Retrieval-Augmented LLMs for Knowledge Graph Completion across General and Biomedical Domains. *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 16432–16445.
   Role: Contributed to the retrieval-augmented framework design and experiments.
4. Feiqing Huang, Jue Hou, Ningxuan Zhou, Kimberly Greco, Chenyu Lin, Sara Morini Sweet, Jun Wen, Lechen Shen, Nicolas Gonzalez, **Sinian Zhang**, et al. Advancing the Use of Longitudinal Electronic

Health Records: Tutorial for Uncovering Real-World Evidence in Chronic Disease Outcomes. *Journal of Medical Internet Research* 27 (2025): e71873.
Role: Contributed the longitudinal EHR data processing and curation.

5. Yongkang Xiao, **Sinian Zhang**, Huixue Zhou, Mingchen Li, Han Yang, Rui Zhang. FuseLinker: Leveraging LLM's Pre-trained Text Embeddings and Domain Knowledge to Enhance GNN-based Link Prediction on Biomedical Knowledge Graphs. *Journal of Biomedical Informatics* 158 (2024): 104730.
Role: Contributed to the GNN link-prediction methodology and experiments.
6. Jue Hou, Rachel Zhao, Jessica Gronsbell, Yucong Lin, Clara-Lea Bonzel, Qingyi Zeng, **Sinian Zhang**, et al. Generate Analysis-Ready Data for Real-world Evidence: Tutorial for Harnessing Electronic Health Records With Advanced Informatic Technologies. *Journal of Medical Internet Research* 25 (2023): e45662.
Role: Contributed to EHR data processing for generating analysis-ready datasets.
7. Shiyao Huang, Junliang Lyu, **Sinian Zhang**, Ruiying Tang, Huan Xiao, Yuanyuan Zhang, Xiaoling Lu. A Post-processing Machine Learning for Activity Recognition Challenge with OpenStreetMap Data. *Adjunct Proceedings of UbiComp/ISWC 2023*, pages 557–562.
Role: Contributed to the post-processing machine-learning method.

Preprints & Under Review

1. Jiajun Liang, Yue Liu, Doudou Zhou, **Sinian Zhang**, Junwei Lu. The Wreaths of KHAN: Uniform Graph Feature Selection with False Discovery Rate Control. *In revision*, *Biometrika*. arXiv preprint: 2403.12284, 2024.
2. Gaoxiang Luo, Yifan Wu, **Sinian Zhang**, Aryan Deshwal, Ju Sun. Aligning Language Models with Selective Prediction. *Under review*, 2026. arXiv preprint: 2607.03528, 2026.
3. **Sinian Zhang***, Chongwei Chen*, Guanchen Li, Ju Sun. Ensemble Selective Classification. *Under review*, 2026.
4. Shruthi Venkatesh*, **Sinian Zhang***, Wen Zhu, Michele Morris, Rocco Mercurio, et al. Predicting the Timing of First Sustained Cognitive Worsening in Alzheimer's Disease Using Real-World Clinical Data and Machine Learning. *Under review*. *medRxiv* preprint: 10.64898/2026.06.02.26354764, 2026.
5. **Sinian Zhang***, Kaicheng Zhang*, Ziping Xu, Tianxi Cai, Doudou Zhou. Generalized Linear Markov Decision Process. arXiv preprint: 2506.00818, 2025.

TECHNICAL SKILLS

Languages: R, Python, C, C++, SQL, LaTeX
Deep Learning: PyTorch, Transformers, LangGraph, LLM APIs (Claude, Gemini, Ollama)
Tools: HPC/Slurm, Git, R package development (CRAN)

SOFTWARE

MultiViewAgent – Agentic AI system (Python, LangGraph) — Private, 2026
Multi-agent workflow that automates multiview biomedical data analysis over the Safo Lab multiview toolkit. Source kept private pending grant submission and a potential patent application.

IntegMultiReg – R package (MCMC sampler in C) — Submitted to CRAN, 2026
Integrative Bayesian multi-regression of multi-platform biomarkers, supporting continuous, binary and survival outcomes with missing platforms.

Khan – R package — 2026
Sole author and maintainer. Companion package to The Wreaths of KHAN: FDR-controlled inference on subgraph features and persistent homology generators in Gaussian graphical and Ising models.

WaTL – R implementation 2025

Reference implementation of Wasserstein Transfer Learning (NeurIPS 2025), including the simulation and real-data analyses.

EnsembleSC – Python implementation 2025

Reference implementation of Ensemble Selective Classification (under review, NeurIPS 2026).

NILE – Python and R implementations 2025

Natural language processing package for clinical narrative extraction; development guided as a mentor.

TALKS & PRESENTATIONS

Conference Abstracts

1. Shruthi Venkatesh, **Sinian Zhang**, Oscar Lopez, Tianxi Cai, Jue Hou, Zongqi Xia. Predicting the Timing of Cognitive Decline in Alzheimer's Disease Using Real-world Clinical Data and Machine Learning (P10-12.009). *Neurology* 106 (11_Supplement_1), 2026.

Talks

1. **Federated Adaptive Causal Estimation with High-Dimensional Covariates (FACE-HD)**
   7th ICSA-Canada Chapter Symposium, McGill University, Montreal, Canada Aug 2026

TEACHING & MENTORING

Shuheng Kong & Conglin Ruan – Master's Students May 2025 – Sept 2025

Guided development of Python and R versions of the NILE package for natural language processing.

Teaching Assistant, Data Science Practice *Renmin University of China*
Beijing, China Jan 2024 – Jun 2024

Teaching Assistant, Probability Theory *Renmin University of China*
Beijing, China Aug 2023 – Dec 2023

Brittany Wang – High School Student Jun 2023 – Sept 2023

Supervised research using real-world data to support an acute ischemic stroke (AIS) study.

ACADEMIC SERVICE

Conference Reviewer: AAAI 2027, NeurIPS 2026, ICLR 2026
Journal Reviewer: Journal of Intelligent Medicine and Healthcare

HONORS & AWARDS

- **1st Place in the 7th Annual UMN Interdisciplinary Health Data Competition, $3000** – University of Minnesota, Twin Cities, 2026
- **Dean's PhD Scholars Award, $5000** – University of Minnesota, Twin Cities, 2024
- **Scholarship for Academic Excellence** – Renmin University of China, 2023